Software Engineering

Assignment-8

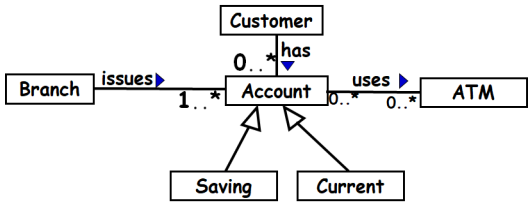
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

For each of the following questions one or more of the given options are correct. Choose the correct options.

QUESTION 1:

Based on the following UML diagram, which of the following statements are correct?



a. A Branch may not have any Account
b. An Account can exist without being issued by a Branch
c. An Account may use multiple ATMs
d. Current inherits from Saving
e. An Account may be issued by 2 Branches
f. A Customer must have at least one Account
g. A Saving is an Account

Correct Answer: c. An account may use multiple ATMs

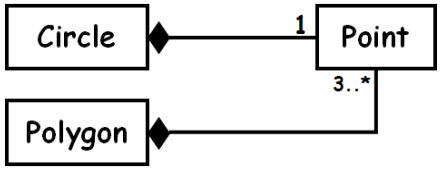
g. A saving is an Account

Detailed Solution:

An account may use multiple ATMs and A saving is an account are the correct option.

QUESTION 2:

Based on the UML relationship shown in the diagram, which of the following statements are correct?



a. A Circle is composed of a Point
b. A Point in the composition relationship has a lifecycle independent of the Circle.
c. The relationship shown is inheritance
d. Polygon and Circle are subclasses of Point
e. The diagram represents a “has-a” relationship between Polygon and Point classes

Correct Answer: a. A circle is a composed of a point

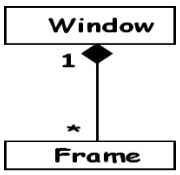
e. The diagram represents a “has-a” relationship between Polygon and Point classes

Detailed Solution:

A Circle is composed of Point(s), indicating a strong whole-part relationship. Polygon has a "has-a" relationship with Point. Composition does not imply inheritance, and Points in composition do not have an independent lifecycle.

QUESTION 3:

Based on the UML composition relationship shown, which of the following statements are correct?



a. A Window may contain multiple Frames

b. A Frame can simultaneously belong to multiple Windows

c. Destruction of a Window implies destruction of its Frames

d. The relationship shown between the classes Window and Frame is composition

e. Frame inherits from Window

Correct Answer: a. A window may contain multiple frames, **c.** Destruction of a window implies destruction of its Frames, **d.** The relationship between the classes Window and Frame is composition

Detailed Solution:

From the above diagram a., c. and d. are the correct option.

QUESTION 4:

Based on the following Java code, which of the following statements are correct?

```
public class Car{
    private Wheel wheels[4];
    public Car (){
        wheels[0] = new Wheel();
        wheels[1] = new Wheel();
        wheels[2] = new Wheel();
         wheels[3] = new Wheel();
    }
}
```

a. A Car object contains exactly four Wheel objects

b. The relationship between Car and Wheel is composition

c. A Wheel object may independently exist outside a Car in this design

d. Car is responsible for creating its Wheel objects

e. Wheel inherits from Car

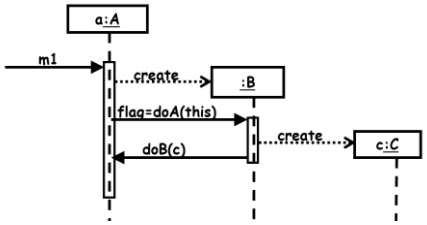
Correct Answer: a. A Car object contains exactly four Wheel objects, **b**. The relationship between car and wheel is composition, **d**. Car is responsible for creating its wheel objects

Detailed Solution:

A Car creates and owns exactly four Wheel objects, indicating a composition relationship. The wheels are created inside the Car constructor.

QUESTION 5:

Based on the UML sequence diagram shown, which of the following statements are correct?



a. The diagram represents interaction among objects over time
b. Dashed vertical lines represent object lifelines
c. Rectangles on lifelines denote object activity/activation
d. Sequence diagrams primarily represent inheritance relationships
e. Objects are typically shown at the top of the diagram

Correct Answer: a. The diagram represents interaction among objects over time**, b.** Dashed vertical lines represent object lifelines, **c**. Rectangles of lifelines denote object activity/activation, **e.** Objects are typically shown at the top of the diagram

Detailed Solution:

Sequence diagram does not represent inheritance relationship.

QUESTION 6:

Based on the comparison between Object-Oriented Analysis (OOA) and Object-Oriented Design (OOD), which of the following statements are correct?

a. OOA focuses on system design
b. OOD is completely independent of implementation constraints
c. OOD refines the analysis model into a design solution
d. OOA is generally implementation-independent
e. OOD takes implementation considerations into account

Correct Answer: c. OOD refines the analysis model into a design solution**, d.** OOA is generally implementation-independent, **e.** OOD takes implementation considerations into account

Detailed Solution:

OOA focuses on analyzing requirements and is mostly implementation-independent. OOD refines the analysis model into an implementable design considering implementation constraints.

QUESTION 7:

In object-oriented analysis, which of the following are typical responsibilities of Boundary objects?

a. Managing persistent business data
b. Interacting with external actors/users
c. Handling communication between the system and its environment
d. Coordinating complex control flow logic
e. Representing user interfaces or external interfaces

Correct Answer: b. Interacting with external actors/users , **c.** Handling communication between the system and its environment, **e.** Representing user interfaces or external interfaces

Detailed Solution:

Boundary objects interact with users/external systems and represent interfaces such as screens, forms, or APIs. They do not store persistent data or control workflow logic.

QUESTION 8:

Which of the following statements about Entity Objects are correct?

a. Entity objects typically store information for transient use
b. Entity objects are mainly responsible for user interaction
c. Entity objects may perform elementary operations such as searching and sorting
d. Entity objects are often identified from nouns in the problem description
e. Entity objects usually coordinate overall system control flow
f. Examples of entity objects include Book and BookRegister

Correct Answer: c. Entity objects may perform elementary operations such as searching and sorting, **d**. Entity objects are often identified from nouns in the problem description, **f.** Examples of entity objects include Book and BookRegister

Detailed Solution:

Entity objects represent persistent business data and may perform simple operations on that data. They are often identified from nouns in the problem domain.

QUESTION 9:

Which one of the following statement about Controller Objects is **FALSE**?

a. Controller objects help realize use case behavior

b. Controller objects interact with boundary objects

c. Controller objects coordinate the activities of entity objects

d. Controller objects primarily store long-term persistent data

e. A single use case may involve multiple controller objects

f. Controller objects often contain significant business logic

Correct Answer: d. Controller objects primarily store long-term persistent data

Detailed Solution:

Controller objects coordinate use-case flow and business logic, but persistent long-term data is mainly handled by Entity objects.

QUESTION 10:

Which of the following statements about composite states in UML state machines are correct?

a. A composite state needs to contain three or more nested substates

b. Composite states help reduce complexity in state diagrams

c. A system can be in only the composite state and never in its substates

d. Composite states support hierarchical state modeling

e. Entry into a composite state may automatically activate one of its substates

Correct Answer: b. Composite states help reduce complexity in state diagrams, **d.** Composite states support hierarchical state modeling, **e.** Entry into a composite state may automatically activate one of its substates

Detailed Solution:

Composite states simplify complex state diagrams and support hierarchical state modeling. Entering a composite state can automatically activate a default substate.

**********END**********